

A voltage source $\underline{E} = 10e^{j\frac{\pi}{3}}$ [...] powers a circuit, determining a current flow through this one of value $\underline{I} = \frac{5\sqrt{3}}{2} + \frac{5}{2}j$ [...]. Compute the complex power of this circuit, $\underline{S}$, and explain the significance of the real part and the imaginary part of this power, and state their measurements units. Find the sinusoidal expression of the current $i(t)$, and compute its value when $t=6.5[\text{ms}]$ if the circuit works on the frequency $f=30[\text{Hz}]$. One circuit element is a resistance. Compute its value starting from the active power. State the measurement units for the following quantities: $\bar{B}$ and ε_0 .

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1. Volt, V - 0,8
2. Ampere, A - 0,8
3. $\underline{S} = \underline{E} \cdot \underline{I}^* = 10e^{j\frac{\pi}{3}} \cdot (\frac{5\sqrt{3}}{2} - \frac{5}{2}j) = 10e^{j\frac{\pi}{3}} \cdot 5e^{-j\frac{\pi}{6}} = 50e^{j(\frac{\pi}{3}-\frac{\pi}{6})} = 50e^{j\frac{\pi}{6}} = 25\sqrt{3} + 25j - 3$
4. P - active power - 2
5. Q - reactive power - 2
6. W (Watt) and VAr (Volt-Ampere reactive) - 1,6
7. $i(t) = 5\sqrt{2} \sin(\omega t - \frac{\pi}{6}) - 2,4$
8. $i(6,5ms) = 5\sqrt{2} \sin(2\pi \cdot 30 \cdot 6,5 \cdot 10^{-3} - \frac{\pi}{6}) \cong 7,05 \sin(\pi \cdot 0,22) \cong 4,55[V] - 3$
9. Ω (Ohm) - 0,8
10. $P = R \cdot I^2 = 25\sqrt{3} \rightarrow R = \frac{P}{I^2} = \frac{25\sqrt{3}}{5^2} = \sqrt{3} = 1,73[ohm] - 2$
11. and 12. T (Tesla) and F/m (Farad/meter) - 1,6

